# EATA: Distribution-Aware Symbolic Discovery for Explainable Trading

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**Abstract**

Profit-seeking yet explainable trading remains elusive: deep neural predictors excel at accuracy but lack regulatory transparency, whereas symbolic regression yields readable formulas that optimize proxy objectives misaligned with economic utility. We present EATA (Explainable Algorithmic Trading Agent), a distribution-aware, neural-guided symbolic regression framework that aligns the objective, hypothesis space, and training signal with trading profitability. EATA couples a three-headed policy–value–profit network (PVPNet) with Monte Carlo Tree Search, where accuracy rewards stabilize search while a Wasserstein-derived profit reward captures asymmetric tail risk. A finance-grounded, decoupled grammar with bounded module libraries encodes moving-average, momentum, RSI, volatility, and conditional operators to curb hypothesis-space inflation, while sliding-window inheritance and quantile-consensus signals enable temporal adaptation. Across representative S&P 500 constituent stocks over 2013–2018, EATA delivers an annualized return of 21.2%, a Sharpe ratio of 0.76, and a maximum drawdown of 17%, outperforming strong baselines while maintaining full interpretability; Pareto-frontier analyses confirm a favorable complexity–performance trade-off. Ablation and objective studies show that removing the Wasserstein rewards, neural guidance, or grammar augmentation severely degrades performance, demonstrating that profit-aware symbolic regression can produce explainable trading strategies competitive with black-box counterparts.

*Keywords:* Algorithmic Trading, Distribution-Aware Learning, Symbolic Regression, Wasserstein Distance, Explainable AI, Optimal Transport, Risk-Sensitive Learning

## 1. Introduction

Algorithmic trading operates at the tense intersection of two competing requirements: although modern deep learning and reinforcement learning systems have achieved state-of-the-art prediction accuracy [1, 2], they lack transparent inductive biases. This opacity conflicts with emerging regulatory and governance requirements such as MiFID II [3], and complicates risk supervision [4, 5, 6]. Practitioners therefore require not only explainable trading strategies, but also methods that can adapt to shifting market regimes, especially because financial return series are heavy-tailed and non-stationary [7]. The core scientific problem is not merely "how to explain a black box", but *how to discover profit-aligned trading rules whose structure itself constitutes the explanation.*

Existing research offers three incomplete solutions. Black-box DL/RL models achieve strong raw performance but provide almost no transparency or safety guarantees. Classic symbolic regression and alpha-mining pipelines can return readable formulas [8, 9], but they typically optimize proxy objectives disconnected from economic returns (MSE, IC), which amplifies multiple-testing bias and turnover costs. Hybrid grammar-guided systems (e.g., AlphaCFG [10], NEMoTS [11]) constrain the search space, yet still inherit point-wise loss objectives and lack mechanisms for temporal adaptation. These limitations motivate our design of a profit-aware, grammar-disciplined discovery agent.

We therefore raise the core research question: *Can we design a symbolic regression agent whose goals, hypothesis space, and training signals are aligned with trading profitability under regulatory and regime constraints?* This question can be broken down into three sub-problems:

- RQ1 **Objective alignment.** Which distributional divergence can better capture asymmetric tail risk than point-wise losses, and how should it regularize the search process without overfitting noise?
- RQ2 **Search discipline.** How can neural priors, grammar constraints, and inheritance mechanisms be integrated to maintain auditability and adapt to market shifts while exploring large-scale symbol spaces?
- RQ3 **Empirical closure.** Considering transaction costs and the S&P 500 universe, can such an agent obtain risk-adjusted returns equivalent to or better than those of black-box baselines?

To answer these questions, we introduce three interre-

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lated innovations: (i) a Wasserstein-guided policy-value-profit network (PVPNet), which decouples the accuracy reward $r^{\text{acc}}$ from the profit-aware reward $r^{\text{rl}}$ and injects the latter into MCTS backpropagation; (ii) a finance-grounded decoupled grammar, including discrete window/lag nonterminals, conditional logic, and a bounded module library to curb hypothesis-space inflation; (iii) a sliding-window pipeline with expression inheritance, module extraction, and quantile-consensus signals to achieve temporal adaptation.

Our contributions are as follows:

1. **Profit-aware symbolic regression.** We formulate the discovery of trading rules as a distribution-matching problem via the 1-Wasserstein distance, and couple profit rewards with accuracy stabilizers to control tail risk under heavy-tailed returns.
2. **Neural-guided grammar search with temporal adaptation.** We integrate a three-headed PVPNet with grammar constraints, module libraries, and sliding-window inheritance mechanisms to explore hypothesis spaces that are both expressive and auditable under regime shifts.
3. **Comprehensive empirical validation.** We provide multi-year, multi-asset experiments, transaction-cost robustness analysis, and ablations that directly address RQ1–RQ3 and demonstrate competitiveness with black-box baselines.

Section 2 situates EATA within the literature on evaluation protocols, symbolic discovery, and risk-sensitive learning; Section 3 elaborates the framework; Section 4 presents empirical evidence; Section 5 discusses limitations and future work.

## 2. Related Work

This section places EATA at the intersection of three areas: (i) empirical algorithmic trading evaluation, (ii) symbolic regression and neural-guided search, and (iii) distributional / risk-sensitive objectives. We organize the existing work according to the pragmatic taxonomy aligned with the research questions: evaluation and backtesting protocols, symbolic discovery and search algorithms, and distributional metrics and risk-sensitive learning.

### 2.1. Evaluation, Backtesting, and Statistical Validity in Trading

Empirical algorithmic trading research typically reports risk-adjusted performance metrics derived from classical portfolio theory [12, 13]. However, in finite samples, the estimated Sharpe ratio may exhibit considerable uncertainty and non-Gaussian sensitivity [14, 15], and naive benchmark selection can trigger multiple-testing and data-snooping biases. Classical econometric methods, such as the reality check [16] and the superior predictive ability (SPA) test [17], provide principled benchmarks for large-scale strategy comparison. Subsequent work further emphasizes the need to correct for selection bias and backtest overfitting [18, 19]. In the broader asset pricing literature, multiple-testing concerns have been extensively documented when factor libraries are mined at scale [20]. In addition to statistical validity, realistic transaction cost modeling is critical, as many strategies see their performance degrade sharply once implementation frictions are introduced [21]. These considerations motivate our emphasis on risk-adjusted metrics and robustness checks, rather than forecast accuracy alone.

A recurring pitfall in the trading literature is equating in-sample predictive performance with out-of-sample trading profitability. This misalignment is further amplified because the evaluation target (returns) itself is noisy, heavy-tailed, and regime-dependent [7]. Consequently, small modeling changes can lead to unstable performance rankings across different assets and time periods.

These challenges motivate EATA's two design principles: (i) reducing implicit multiple testing by grammatically constraining the hypothesis space, and (ii) aligning learning signals with economically meaningful outcomes to avoid proxy-driven improvements.

Finally, we note that algorithmic trading is not solely a modeling problem but also a market-structure problem: automated execution and algorithmic strategies can affect liquidity and price dynamics [22], further motivating our adoption of an evaluation process that emphasizes robustness and auditability.

### 2.2. Interpretable Alpha Discovery and Formulaic Trading Signals

A long-standing research direction focuses on human-interpretable, formulaic trading signals and alpha factors. Technical analysis has been extensively reviewed, but the evidence for its profitability is mixed and shows significant sensitivity to market regimes and evaluation protocols [23]. Recent "alpha mining" work aims to automatically discover and combine formulaic factors at scale, spanning genetic-programming-based alpha discovery [24] to reinforcement learning and language-model-assisted frameworks [25, 26, 27]. Among these, AlphaStock [28] employs an interpretable deep reinforcement learning framework to construct a buying-winners-and-selling-losers portfolio, demonstrating that RL-driven alpha signals can be both profitable and auditable at the portfolio level. These methods reinforce a key methodological insight: interpretability does not automatically imply economic usefulness unless the objective is aligned with trading utility.

Alpha discovery systems typically rely on the assumption that factors improving a proxy objective (e.g., prediction loss, IC, or a simplified RL reward) will translate into profitable and stable trading rules. In practice, this assumption breaks down under at least three common constraints.

1. **Turnover and costs:** Weak forecast signals near zero can trigger frequent trading and fail under realistic friction models [21].
2. **Tail constraints:** Many practitioners operate under drawdown limits; factors that occasionally generate large losses may be unacceptable even if the mean return is positive.
3. **Hypothesis-space inflation:** Large-scale factor libraries increase the likelihood of false discoveries unless the search process is explicitly regularized and the evaluation accounts for multiple testing [16, 20].

EATA can be regarded as a "constrained alpha miner": it retains formulaic interpretability while binding the discovery process to a profit-aware, distribution-sensitive learning signal and using grammar constraints to limit the degrees of freedom.

Consequently, we treat profit-awareness, distribution-sensitivity, and hypothesis-space control (grammar constraints) as first-class design components rather than implementation details.

### 2.3. Black-Box Learning for Trading and the Interpretability Gap

Deep learning and reinforcement learning have been widely applied in trading, including recurrent architectures for financial forecasting [1], transformers for multi-scale pattern extraction [29, 30], and deep RL systems for sequential decision making [31, 32]. Modern chronological Transformer variants such as Informer [33], Autoformer [34], and PatchTST [35] have shown improved efficiency in long-sequence financial forecasts; we use a vanilla Transformer as a representative baseline because our focus is on the discovery objective rather than architectural innovation, and our grammar-constrained symbolic expressions operate on pre-computed technical indicators rather than directly on the raw sequence. These models can achieve strong empirical performance, but their inductive biases remain largely opaque and may be highly sensitive to non-stationarity and evaluation protocol choices.

In addition to neural models, tree ensembles remain highly competitive in financial prediction pipelines due to their sample efficiency and robustness to heterogeneous features. Gradient boosting methods such as XGBoost [36] and LightGBM [37] are widely used as baselines in practice. However, even when post-hoc explanation tools are applied, ensemble models can produce unstable interpretations across correlated features or under regime shifts. Rule extraction methods such as inTrees [38] partially bridge this gap, but they typically explain a fixed predictor rather than directly optimizing the discovery process for interpretable and deployable trading rules.

The black-box literature typically assumes that improvements in predictive capacity will translate into improved trading performance under deployment constraints. However, this assumption is particularly fragile in finance because: (i) low signal-to-noise ratios amplify overfitting, (ii) regime shifts destroy stationarity, and (iii) decision-making must satisfy auditability and compliance requirements [3, 5, 6]. Furthermore, although post-hoc explanation methods (LIME/SHAP) can provide local rationales, they may not meet robustness and faithfulness requirements and can even be manipulated [39, 40, 41]. This motivates our adoption of intrinsically interpretable strategies, where the decision rules themselves constitute the model.

### 2.4. Symbolic Regression for Time Series: Evolutionary, Grammar-Based, and Neural

**Evolutionary and Grammar-Based GP.** Genetic Programming (GP) remains the foundational paradigm for symbolic regression [8, 42]. Grammar-based GP provides a principled way to encode domain constraints and reduce invalid expressions [43]. Modern toolkits such as gplearn [44] and Operon [45] improve scalability through efficient evaluation. In computational finance, evolutionary methods have been widely explored for alpha discovery and rule mining [46], but they often suffer from bloat and expensive fitness evaluations, particularly when the objective function requires actual backtesting.

**Neural Symbolic Regression.** Recent studies have introduced neural guidance to improve sample efficiency and scalability. DSR [47] uses policy gradients to generate expressions token-by-token. More recently, transformer-based neural-to-symbolic models have emerged as a powerful paradigm: [48] proposed a transformer architecture that learns to generate symbolic expressions directly from numeric data in an end-to-end fashion, while [49] extended this approach by training transformers to directly output symbolic expression trees, achieving strong generalization on out-of-distribution equations. [50] further demonstrated that pre-trained language models can be fine-tuned for symbolic regression tasks. Despite these advances, transformer-based methods typically require large training corpora of synthetic equations and may struggle to generalize when the underlying functional forms differ substantially from the training distribution—a particular concern in financial markets where the true data-generating process is unknown and non-stationary. A comprehensive review [51] emphasizes that designing objective functions reflecting the downstream utility of discovered expressions—rather than mere reconstruction error—remains a core challenge shared across all symbolic regression paradigms.

In addition to direct expression generation, grammar-aware generative models offer another avenue for imposing syntactic constraints while learning expressive priors. For example, grammar variational autoencoders [52] demonstrate how context-free grammars can ensure syntactic validity. Furthermore, program synthesis and library learning methods such as DreamCoder [53] learn reusable subroutines to improve sample efficiency and compress the hypothesis space. These ideas are conceptually consistent with EATA's module extraction: rather than learning an implicit program prior, we explicitly maintain a bounded library of reusable sub-expressions for auditability.

For time-series trading, two practical constraints are often overlooked.

1. **Evaluation-dominated cost:** In trading, the expensive step is not parsing the expressions but evaluating them under the backtesting protocol (positions, costs, and risk metrics). Therefore, methods that generate a large number of candidates without strong pruning are computationally infeasible.
2. **Spuriousness under non-stationarity:** Without clear domain constraints, symbolic regression may fit transient correlations that vanish under regime shifts; even if the expressions are interpretable, they may be unstable.

As a structural prior, the grammar constraints in EATA serve both scientific and practical purposes: scientifically, they make domain knowledge operational and limit the hypothesis class; practically, they reduce the scale of the multiple-testing problem induced by search.

We explicitly encode financial operators and fixed horizons to balance expressiveness against robustness and auditability.

### *2.5. Neural-Guided Search and MCTS for Symbolic Discovery*

Monte Carlo Tree Search (MCTS) [54, 55] provides a principled exploration–exploitation trade-off and has been successfully combined with neural policy–value guidance in game-playing systems [56, 57]. In symbolic regression, MCTS-based methods have been explored as alternatives to genetic programming [58, 59], with actor–critic style enhancements proposed to improve search efficiency [60]. More recently, Huang et al. [61] enhanced MCTS for symbolic regression through improved exploration strategies. The most relevant to our work is NEMoTS (Neural Enhancement Monte Carlo Tree Search) [11], which pairs MCTS with policy–value networks for time-series symbolic regression. Similarly, AlphaCFG [10] treats alpha discovery as a grammar-guided generation problem, using a tree-structured linguistic MDP to enforce syntactic validity. RiskMiner [62] formalizes alpha mining as a risk-seeking MCTS problem on tokenized expressions, demonstrating that MCTS-based search can scale to large financial feature spaces. EATA shares the neural-guided search paradigm but differs by explicitly steering the guidance signal toward profit-aware objectives (via Wasserstein distance) rather than optimizing IC alone, and by introducing a temporal adaptation mechanism.

More broadly, MCTS has been used as a general mechanism for symbolic and scientific discovery beyond time series, such as discovering governing equations [63]. This line of work highlights an important methodological point: search algorithms benefit from domain priors and stable evaluation signals; otherwise, they risk wasting most of their budget exploring syntactically valid but semantically uninformative regions.

Existing MCTS-based symbolic regression methods typically optimize prediction fit or benchmark-task accuracy; the resulting expressions may be interpretable but are economically weak when deployed as trading rules. Specifically, optimizing MSE can yield accurate predictions near the mean, but the calibration degrades in the tails — precisely where trading risks are concentrated.

Two additional boundary conditions are important in financial markets.

1. **Regime dependence:** Expressions discovered in one time window may decay when volatility, liquidity, or macroeconomic conditions change.
2. **Objective mismatch:** Even if a symbolic model predicts returns, a decision layer is needed to convert predictions into trading rules; if the objective ignores this decision-making layer, the downstream strategy may be unstable.

EATA addresses these challenges by coupling neural-guided MCTS with a profit-aware head and employing a sliding-window inheritance mechanism to steer the search toward reusable substructures that persist across regimes.

### *2.6. Distributional Metrics, Optimal Transport, and Risk-Sensitive Learning*

Financial returns are heavy-tailed and asymmetric [7], making point-wise regression losses (e.g., MSE) a weak proxy for economic utility. Optimal transport provides a geometrically meaningful way to compare distributions, as systematically developed in the foundational literature and modern computational treatments [64, 65]. Recent work has applied Wasserstein distributionally robust optimization (DRO) to portfolio selection, proving that out-of-sample stability can be achieved when the return distribution lies within a Wasserstein ball around the empirical measure [66, 67]. In reinforcement learning, distributional perspectives [68] and risk-sensitive objectives (e.g., CVaR optimization [69]) further promote designs that reflect tail risk and distributional shape rather than mean outcomes alone. Kastner et al. [70] formalized distributional model equivalence in risk-sensitive RL and established theoretical connections with modern portfolio theory. Recent work on exponential-criterion risk-sensitive RL [71] and risk-preference-adaptive distributional RL [72] for stock trading further demonstrates the importance of tail-risk-aware objectives in financial decision-making. Zhang [73] proposed tail-safe hedging strategies that explicitly account for extreme events. These insights directly motivate our use of the Wasserstein distance as a profit-aware training signal.

Although risk-sensitive criteria (e.g., CVaR) directly target tail losses, they typically require careful adjustment of tail parameters and may be unstable when tail events are rare. The Wasserstein distance provides a complementary mechanism: it compares the entire distribution and remains meaningful even when the supports differ.

The key design question is not "which metric is more elegant in theory" but "which metric provides a stable learning signal in heavy-tailed samples." In this sense, the Wasserstein distance achieves a distribution-level inductive bias: it penalizes systematic quantile miscalibration, thereby suppressing strategies that perform well on average but fail in tail events.

### 2.7. Explainability, Robustness, and Regulation in Finance

In high-risk domains such as finance, interpretability is closely tied to governance and compliance requirements, including regulatory constraints (e.g., MiFID II [3]) and broader discussions on the right to explanation [74]. Although post-hoc explanation methods such as LIME [39] and SHAP [40] are widely used, their robustness and faithfulness have been questioned, particularly under adversarial manipulation [41]. Furthermore, there is ongoing debate about whether attention mechanisms constitute valid explanations [75, 76, 77]. These findings support the view that intrinsically interpretable symbolic strategies may be preferable in terms of auditability and scientific scrutiny [4, 5, 6].

The core tension lies between *post-hoc transparency* and *intrinsic interpretability*. Post-hoc explanations can provide local insights but may not satisfy robustness or auditability requirements; symbolic strategies, by contrast, directly expose the decision rule but risk overfitting unless the hypothesis space and objective functions are carefully designed. EATA's design treats interpretability as a constraint on the hypothesis class (via grammar), rather than as a visualization layer applied after training.

An emerging complementary direction is designing models that are self-explaining by construction (rather than interpreted post-hoc), such as self-explaining neural networks [78]. However, such methods still require careful definition of what constitutes a faithful interpretation and may not produce globally auditable rule sets. In contrast, symbolic expressions provide explicit, globally applicable decision rules that can be inspected and stress-tested.

### 2.8. Positioning of EATA

Table 1: Comparison with Related Approaches

| Method | Interp. | Neural | Finance | Profit | Regime Adapt. |
|---|---|---|---|---|---|
| LSTM/Transformer | ✗ | ✓ | ✗ | ✗ | ✓ |
| FinRL (PPO) | ✗ | ✓ | ✓ | ✓ | ✓ |
| GP (Symbolic) | ✓ | ✗ | ✗ | ✗ | ✗ |
| Alpha Mining [25, 27] | ✓ | ✓ | ✓ | ✓ | ✗ |
| DSR | ✓ | ✓ | ✗ | ✗ | ✗ |
| NEMoTS | ✓ | ✓ | ✗ | ✗ | ✗ |
| **EATA (Ours)** | ✓ | ✓ | ✓ | ✓ | ✓ |

Table 1 compares EATA with representative method families across key dimensions: interpretability (Interp.), use of neural components (Neural), domain-specific financial operators (Finance), profit-aware objective (Profit), and explicit mechanisms for non-stationarity adaptation (Regime Adapt.). Compared to prior alpha mining and symbolic discovery pipelines, EATA uniquely integrates *neural-guided search* (MCTS + PVPNet), *financial operators* (domain grammar), and *profit-aware learning* (distributional reward) within a unified, interpretable framework.

The above literature reveals that the core difficulty lies neither in the lack of interpretable models nor in the lack of strong search capabilities, but rather in the large-scale, implicit multiple testing of the hypothesis space and the *mismatch between discovery objectives and trading utility.* EATA's components are directly designed to address these failure modes: the Wasserstein-based profit signal targets distribution-level utility under heavy tails; grammar constraints limit the degrees of freedom and improve auditability; neural-guided MCTS improves search efficiency while maintaining explicit exploration; and sliding-window inheritance provides a practical mechanism for responding to non-stationarity.

## 3. Methodology

This section presents the technical details of the Explainable Algorithmic Trading Agent (EATA). We first establish a unified notation system and then describe the core algorithm and its design principles.

### 3.1. Problem Formulation

Given a multivariate financial time series $\mathbf{X} = \{\mathbf{x}_t\}_{t=1}^{T}$ where $\mathbf{x}_t \in \mathbb{R}^d$ represents $d$ market features at time $t$, we seek an interpretable expression $e^*$ that balances predictive accuracy with trading profitability:

$$e^* = \arg\max_{e \in \mathcal{E}} \left[\eta \cdot \mathcal{A}(e, \mathbf{X}) + (1-\eta) \cdot \mathcal{P}(e, \mathbf{X})\right], \quad (1)$$

where $\mathcal{E}$ denotes the valid expression space defined by grammar $\mathcal{G}$, $\mathcal{A}(\cdot)$ measures prediction accuracy, $\mathcal{P}(\cdot)$ measures trading profit, and $\eta \in [0, 1]$ balances the two objectives.

A pure accuracy objective is inconsistent with trading utility because it treats all errors symmetrically and ignores the asymmetry of risk. Conversely, a pure profit objective may overfit to specific return realizations and produce brittle expressions that exploit idiosyncratic fluctuations. The mixed objective in Eq. 1 acts as a regularizer: $\mathcal{A}$ stabilizes the search toward expressions that capture repeatable predictive structure, while $\mathcal{P}$ biases discovered expressions toward economically significant outcomes. The control parameter $\eta$ thus provides a transparent trade-off between statistical fit and economic utility, which can be adjusted according to risk preferences.

### 3.2. Domain-Specific Grammar Design

We employ a context-free grammar $\mathcal{G} = (\mathcal{V}, \Sigma, \mathcal{R}, S)$ tailored for financial time series analysis. The grammar incorporates domain knowledge through fixed-window financial operators.

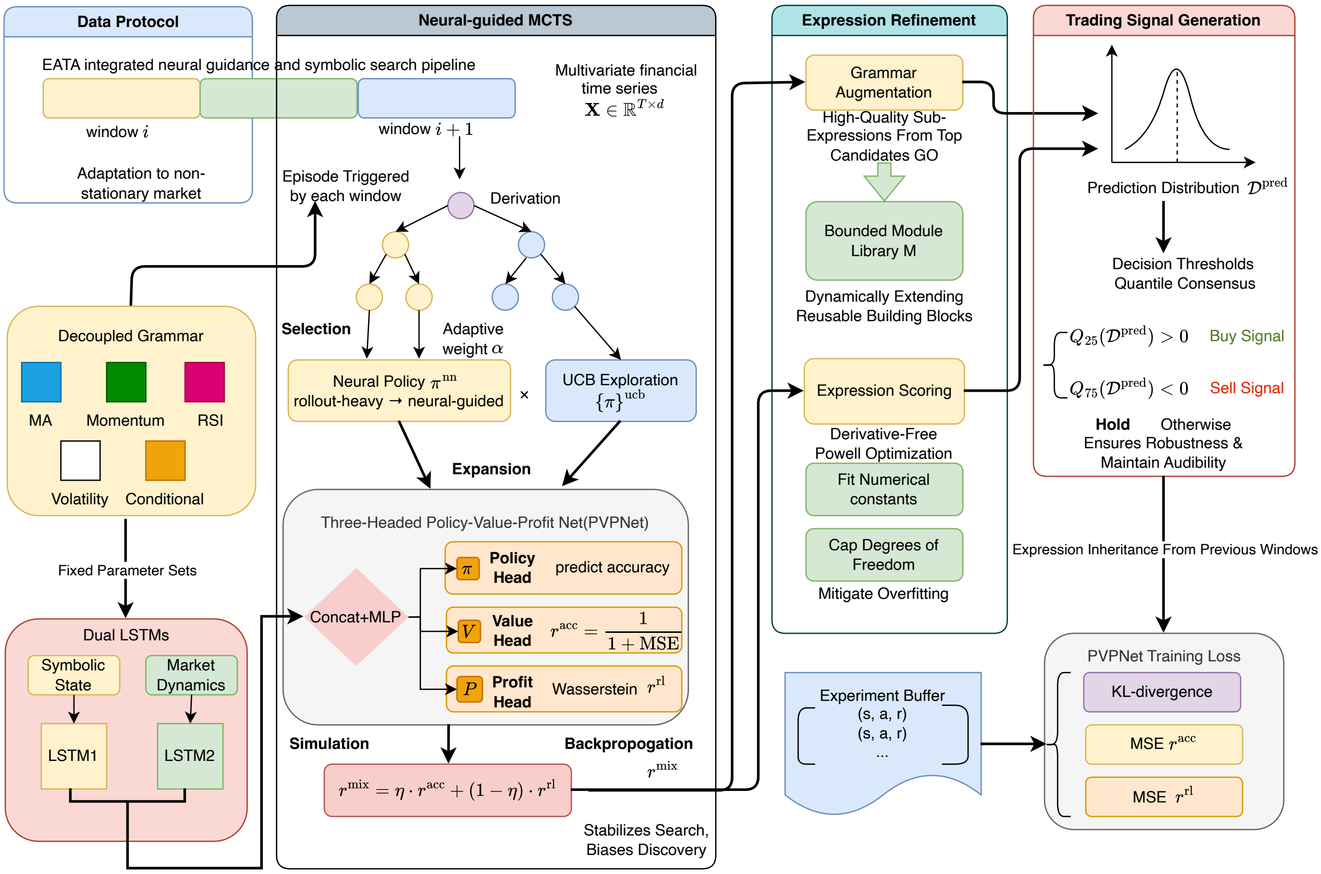



Figure 1: EATA Integrated Neural Guidance and Symbolic Search Pipeline. The framework operates through four main components: (1) **Data Protocol**: Sliding window adaptation with decoupled grammar (MA, Momentum, RSI, Volatility, Conditional operators); (2) **Neural-Guided MCTS**: Episode-triggered derivation with selection, expansion, and adaptive weight $\alpha$ balancing neural policy $\pi_{nn}$ and UCB exploration $\pi_{ucb}$; (3) **Expression Refinement**: Grammar augmentation via bounded module library, expression scoring with derivative-free Powell optimization, and numerical constant fitting; (4) **Trading Signal Generation**: Three-headed PVPNet (Policy-Value-Profit) with dual LSTMs encoding symbolic states and market dynamics, profit-aware reward via 1-Wasserstein distance aligning predicted and actual return distributions, and quantile-based consensus signals (Q25/Q75 thresholds for Buy/Sell decisions).

**Non-terminals and Terminals.** The non-terminal set $\mathcal{V} = \{S, A, B, D\}$ includes the start symbol ($S$), primary expressions ($A$), helper expressions ($B$), and decision predicates ($D$). The terminal set $\Sigma$ comprises feature variables $\{x_0, x_1, \dots, x_{d-1}\}$, constants, and operators.

**Production Rules.** The rule set $\mathcal{R}$ comprises the start rule and four operator categories, including:

1) Start Rule

$$S \to A. \tag{2}$$

2) Basic Arithmetic

$$\begin{aligned} A \to\ & A + A \mid A - A \mid A \times A \mid A \div A \\ & \mid A \times C \mid A + C. \end{aligned} \tag{3}$$

3) Mathematical Functions

$$\begin{aligned} A \to\ & \cos(A) \mid \sin(A) \mid \exp(A) \mid \log(B) \mid \sqrt{B} \\ & \mid |A| \mid A^2. \end{aligned} \tag{4}$$

4) Financial Operators (Decoupled Parameters):

$$\begin{aligned} A \to\ & \mathrm{delay}(A, L) \mid \mathrm{ma}(A, W) \mid \mathrm{diff}(A, L) \mid \mathrm{mom}(A, L) \\ & \mid \mathrm{max\_n}(A, W) \mid \mathrm{min\_n}(A, W) \mid \mathrm{rsi}(A, W) \mid \mathrm{vol}(A, W) \end{aligned} \tag{5}$$

$$W \to 5 \mid 10 \mid 20 \mid 30 \mid 60, \quad L \to 1 \mid 5 \mid 10 \mid 20. \tag{6}$$

5) Conditional Logic

$$A \to \mathrm{ite}(D, A, A), \quad D \to A > B \mid A < B. \tag{7}$$

6) Helper Expressions (Guaranteed Non-Negative Output):

$$\begin{aligned} B \to\ & |A| \mid A^2 \mid \mathrm{ma}(A, W) \mid \mathrm{vol}(A, W) \\ & \mid \mathrm{max\_n}(A, W) \mid \mathrm{min\_n}(A, W). \end{aligned} \tag{8}$$

We adopt a decoupled grammar design inspired by AlphaCFG [10], separating operators from their parameters. This introduces parameter non-terminals ($W$ for windows, $L$ for lags) that resolve to a discrete set of valid values. This compositional structure reduces the number of unique production rules while maintaining expressiveness, and prevents the generation of semantically invalid parameter combinations (e.g., negative windows). The conditional operator ite enables piecewise expressions for regime-switching strategies.

In finance practise, unconstrained operator sets often produce expressions that are either uninterpretable (deeply nested non-linearities) or non-robust (operators search for hyperparameters via continuous constants). Fixed-window operators serve two purposes: they encode domain priors and reduce multiple-testing effects by limiting the degrees of freedom. In contrast, allowing arbitrary window lengths would significantly expand the hypothesis space and increase the risk of data snooping [16, 18]. Grammar constraints can thus be viewed as a structural regularizer that improves both interpretability and generalization.

### *3.3. Neural-Guided Monte Carlo Tree Search*

Algorithm 1 presents the MCTS procedure with neural network guidance. Each episode traverses the search tree using a policy fusion mechanism.

The dual fusion mechanism addresses the uncertainty of neural network predictions in the early stages of training. When $\alpha$ is small (the buffer is not yet full), the rollout provides reliable value estimates; as training progresses, $\alpha$ increases and the reliance on the neural network grows accordingly. The value-profit fusion via $\omega$ ensures that selected expressions account for both predictive fit and profit potential.

**Why MCTS (vs. GP / sequence generators)?** GP is effective but suffers from bloat and high evaluation costs, especially with backtesting-based objectives. Sequence generators (RNN/Transformer) scale well but typically require large amounts of training data and may produce syntactically valid but semantically meaningless expressions without strong constraints. MCTS provides a controllable exploration mechanism: it leverages neural priors when available while retaining explicit exploration via UCB. This is especially suitable for financial time series, where the signal-to-noise ratio is low and naive exploitation can lead to overfitting.

### *3.4. Expression Scoring with Coefficient Optimization*

Algorithm 2 evaluates expressions and optimizes numerical constants.

Constant fitting is indispensable, because grammar encodes the functional forms, and the numerical coefficients determine the scale and sensitivity. If there is no coefficient optimization, the search will either discard the correctly structured expression because of poor initialization constants, or expand the grammar to contain many approximately repeated scaling variants, thus increasing the hypothesis space. We use a derivative-free optimizer (Powell) because expression evaluation may be non-smooth (for example, due to the existence of conditional operators) and the gradient is not easy to obtain. The constant cap

**Algorithm 1** Neural-Guided MCTS for Symbolic Regression

**Require:** Data $\mathbf{X}$, grammar $\mathcal{G}$, network $f_\theta$, episodes $K$, guidance weight $\alpha$, fusion weight $\omega$, prior tree $\tau^{\text{prev}}$
**Ensure:** Best expression $e^*$, best tree $\tau^*$, experience buffer $\mathcal{D}$

Initialize $Q(\cdot) \leftarrow 0$, $N(\cdot) \leftarrow 1$, $\mathcal{D} \leftarrow \emptyset$
$s \leftarrow$ InitializeState($\tau^{\text{prev}}$)
**for** $k = 1$ **to** $K$ **do**
  *states* $\leftarrow []$, *actions* $\leftarrow []$
  $\mathcal{U} \leftarrow$ GetUnvisited($s$)
  **while** $\mathcal{U} = \emptyset$ **and** not IsTerminal($s$) **do** ▷ Selection
    $\boldsymbol{\pi}^{\text{nn}}, V, P \leftarrow f_\theta(s, \mathbf{X})$
    $\boldsymbol{\pi}^{\text{ucb}} \leftarrow$ ComputeUCB($s, C$)
    $\boldsymbol{\pi} \leftarrow \alpha \cdot \boldsymbol{\pi}^{\text{nn}} + (1-\alpha) \cdot \boldsymbol{\pi}^{\text{ucb}}$
    $\boldsymbol{\pi} \leftarrow$ Normalize($\boldsymbol{\pi}$)
    $a \sim$ Categorical($\boldsymbol{\pi}$)
    *states*.append($s$), *actions*.append($a$)
    $s \leftarrow$ Step($s, a$)
    $\mathcal{U} \leftarrow$ GetUnvisited($s$)
    **if** $|s| \geq L^{\max}$ **then**
      Backprop(*states*, *actions*, 0)
      **break**
    **end if**
  **end while**
  **if** $\mathcal{U} \neq \emptyset$ **then** ▷ Expansion
    $\boldsymbol{\pi}^{\text{nn}}, V, P \leftarrow f_\theta(s, \mathbf{X})$
    $a \sim$ Categorical($\alpha \cdot \boldsymbol{\pi}^{\text{nn}} + (1-\alpha) \cdot$ Uniform($\mathcal{U}$))
    $s' \leftarrow$ Step($s, a$)
    $r^{\text{acc}}, e \leftarrow$ Evaluate($s', \mathbf{X}$) ▷ Accuracy reward via Alg. 2
    **if** IsTerminal($s'$) **then**
      UpdateModules($e, r^{\text{acc}}$)
    **end if**
  **else** ▷ Simulation via leaf evaluation
    $r^{\text{acc}}, e \leftarrow$ Evaluate($s, \mathbf{X}$) ▷ Accuracy reward via Alg. 2
  **end if**
  $r^{\text{rl}} \leftarrow$ ComputeRLReward($e, \mathbf{X}$) ▷ Profit reward via Eq. 10
  $v^{\text{nn}} \leftarrow \omega \cdot V + (1-\omega) \cdot P$ ▷ Fuse value heads
  $v^{\text{roll}} \leftarrow$ Rollout($s, n^{\text{play}}$) ▷ Random simulations
  $r^{\text{final}} \leftarrow \alpha \cdot v^{\text{nn}} + (1-\alpha) \cdot v^{\text{roll}}$ ▷ Fuse neural and rollout
  Backprop(*states*, *actions*, $r^{\text{final}}$)
  $\mathcal{D}$.append(($\textit{states}, \mathbf{X}, \boldsymbol{\pi}, r^{\text{acc}}, r^{\text{rl}}$))
**end for**
**return** $e^*, \tau^*, \mathcal{D}$

**Algorithm 2** Expression Scoring with Coefficient Estimation

**Require:** Expression $e$, data $(\mathbf{X}, \mathbf{y})$
**Ensure:** Score $r$, optimized expression $e^{\text{opt}}$

$n_c \leftarrow \textsc{CountConstants}(e)$
**if** $n_c = 0$ **then**
    $\hat{\mathbf{y}} \leftarrow \textsc{Evaluate}(e, \mathbf{X})$
**else if** $n_c \geq 10$ **then**
    **return** $0, e$ ▷ Too many constants
**else**
    $\mathbf{c}^* \leftarrow \arg\min_{\mathbf{c}} \|\textsc{Evaluate}(e, \mathbf{X}, \mathbf{c}) - \mathbf{y}\|_2$ ▷ Powell optimization, timeout $t^{\max} = 30\text{s}$
    $e^{\text{opt}} \leftarrow \textsc{Substitute}(e, \mathbf{c}^*)$
    $\hat{\mathbf{y}} \leftarrow \textsc{Evaluate}(e^{\text{opt}}, \mathbf{X})$
**end if**
$\text{MSE} \leftarrow \frac{1}{T}\|\hat{\mathbf{y}} - \mathbf{y}\|_2^2$
$r \leftarrow \frac{1}{1+\text{MSE}}$ ▷ Accuracy reward
**return** $r, e^{\text{opt}}$

alleviates a common failure mode in symbolic regression: degrees of freedom may produce brittle expressions that fit the noisy targets. Other optional methods include linear least squares for the linear-in-parameters forms, or gradient-based optimization for differentiable operators; our choice prioritizes robustness in heterogeneous expression structures.

### *3.5. Grammar Augmentation via Module Extraction*

Algorithm 3 extracts high-quality sub-expressions.

**Algorithm 3** Grammar Augmentation

**Require:** Module library $\mathcal{M}$, expression $e$, reward $r$, max length $L^{\text{mod}}$, max size $M^{\max}$
**Ensure:** Updated library $\mathcal{M}'$

$\mathcal{M}' \leftarrow \mathcal{M}$
**for** each sub-expression $m \in \textsc{Extract}(e)$ **do**
    **if** $\textsc{Length}(m) \leq L^{\text{mod}}$ **and** $m \notin \mathcal{M}'$ **then**
        **if** $|\mathcal{M}'| < M^{\max}$ **then**
            $\mathcal{M}' \leftarrow \mathcal{M}' \cup \{(m, r)\}$
        **else if** $r > \min_{(m',r') \in \mathcal{M}'} r'$ **then**
            $\mathcal{M}' \leftarrow \mathcal{M}' \setminus \{(m^{\min}, r^{\min})\}$
            $\mathcal{M}' \leftarrow \mathcal{M}' \cup \{(m, r)\}$
        **end if**
    **end if**
**end for**
**return** $\mathcal{M}'$

Module extraction is a bias–variance trade-off in symbolic discovery. From the perspective of search, reusing high-quality sub-expressions can shorten the effective depth required to construct excellent candidate expressions, thus improving sample efficiency. From the perspective of generalization, a bounded module library can prevent uncontrolled bloat and limit the number of reusable building blocks, thus reducing the risk of idiosyncratic patterns. Other options include: maintaining a unbounded library (the risk of overfitting and redundancy is higher), using the library learning style of probabilistic program induction, or learning the continuous embeddings of modules; we use explicit, bounded module libraries to maintain interpretability and maintain transparent search space.

### *3.6. Profit-Aware RL Reward via Wasserstein Distance*

To explicitly optimize trading profitability, we introduce a Reinforcement Learning (RL) reward signal $r^{\text{rl}}$. Unlike standard regression targets that penalize point-wise errors (e.g., MSE), financial trading must capture the distributional properties of returns, particularly the asymmetry between gains and losses and the presence of heavy tails. This is consistent with the broader trend in distributional and risk-sensitive reinforcement learning, where methods model return distributions or tail-risk objectives rather than expected value alone [68, 69].

Our key insight is that the Wasserstein distance (Earth Mover's Distance) provides a principled way to measure the return distribution, accounting for both tail risk and directional asymmetry [64, 65]. Unlike KL divergence or MSE, the Wasserstein distance measures the minimum "work" required to transform one distribution into another, making it sensitive to the magnitude and location of distributional differences.

Given predictions $\hat{\mathbf{y}}^{(1)}, \ldots, \hat{\mathbf{y}}^{(k)}$ from the top-$k$ discovered expressions, we construct a prediction distribution $\mathcal{D}^{\text{pred}}$. We compare this against the distribution of actual future returns $\mathcal{D}^{\text{actual}}$ using the 1-Wasserstein distance (Earth Mover's Distance) [64, 65]:

$$\mathcal{W}_1(\mathcal{D}^{\text{pred}}, \mathcal{D}^{\text{actual}}) = \int_{-\infty}^{\infty} |F^{\text{pred}}(x) - F^{\text{actual}}(x)| \, dx, \quad (9)$$

where $F^{\text{pred}}$ and $F^{\text{actual}}$ are the cumulative distribution functions (CDFs) of the predicted and actual returns, respectively. The RL reward is then defined as:

$$r^{\text{rl}} = \frac{1}{1 + \mathcal{W}_1(\mathcal{D}^{\text{pred}}, \mathcal{D}^{\text{actual}})}. \quad (10)$$

Return distributions are inherently non-Gaussian, typically exhibiting skewness and leptokurtosis. For such data, the Wasserstein distance offers significant theoretical advantages over KL divergence or MSE: (1) it provides a meaningful measure even when the supports of the two distributions do not overlap; (2) it respects the geometry of the underlying space (i.e., the magnitude of errors matters, not just differences in probability mass); (3) by measuring the transport cost between the predicted and actual distributions, it effectively captures tail risks [64, 65]. This ensures that the agent is penalized not only for errors in mean prediction, but also for failing to model the asset's risk profile. Beyond these theoretical advantages, recent studies show that Wasserstein DRO improves out-of-sample stability in portfolio selection [66, 67], providing empirical motivation for its use as a trading objective. Appendix C provides a formal analysis of the gradient properties, convergence guarantees, and CVaR-control relationship of the Wasserstein reward.

We emphasize that the superiority of Wasserstein over alternative objectives is not merely empirical but rooted in three structural properties that are uniquely suited to financial return distributions. **First, the quantile uniform-weighting property.** In one dimension, the 1-Wasserstein distance between empirical distributions reduces to $\mathcal{W}_1 = \frac{1}{N}\sum_{i=1}^{N}|x_{(i)} - y_{(i)}|$, the L1 distance between order statistics. This assigns equal weight to every quantile, including the extreme tails, whereas MSE is dominated by the central mass (large squared deviations in the tails are rare and contribute little to the expected MSE). For heavy-tailed returns where extreme events drive realized utility, this difference is decisive: Wasserstein ensures that tail calibration error is penalized proportionally, while MSE is virtually insensitive to tail misspecification. **Second, the dual Kantorovich–Rubinstein representation** $\mathcal{W}_1(P,Q) = \sup_{\|f\|_{\mathrm{Lip}}\le 1} \mathbb{E}_{x\sim P}[f(x)] - \mathbb{E}_{x\sim Q}[f(x)]$ reveals that minimizing Wasserstein distance is equivalent to minimizing distributional mismatch under all 1-Lipschitz test functions simultaneously. This provides a form of adversarial distributional robustness: a strategy that achieves low Wasserstein distance cannot be easily exploited by any smooth payoff function, which is precisely the property needed for out-of-sample robustness in trading. **Third, the CVaR bound** (Proposition 2 in Appendix C) establishes $|\mathrm{CVaR}_\alpha(P) - \mathrm{CVaR}_\alpha(Q)| \le \frac{1}{1-\alpha}\mathcal{W}_1(P,Q)$, showing that minimizing Wasserstein distance directly controls tail-risk discrepancies without the need for explicit CVaR tuning or threshold selection. In contrast, objectives like MSE or KL divergence provide no such finite-sample tail guarantees. Together, these properties explain why Wasserstein-based training yields superior tail-risk control and risk-adjusted returns, as confirmed by our ablation studies in Section 4.

#### *3.6.1. Reward Semantics, Training Targets, and Objective Variants*

EATA uses two conceptually distinct signals during search and training. **(i) Accuracy reward** evaluates point-wise prediction fit, consistent with conventional regression objectives:

$$r^{\mathrm{acc}} = \frac{1}{1 + \mathrm{MSE}}. \tag{11}$$

**(ii) Profit-aware reward** evaluates distributional alignment between predicted and realized return distributions via Wasserstein distance, as defined in Eq. 10.

**Data flow.** During MCTS, each terminal expression is evaluated to obtain $(r^{\mathrm{acc}}, r^{\mathrm{rl}})$; the search backup uses a scalar reward that prioritizes profit while retaining an optional accuracy component:

$$r^{\mathrm{mix}} = \eta \cdot r^{\mathrm{acc}} + (1-\eta)\cdot r^{\mathrm{rl}}. \tag{12}$$

This formulation avoids degenerate expressions that exploit noisy returns: $r^{\mathrm{acc}}$ stabilizes the search toward repeatable predictive structure, while $r^{\mathrm{rl}}$ biases discovery toward economically meaningful tail behavior.

**PVPNet targets.** The value head $V_\theta$ is trained to predict $r^{\mathrm{acc}}$, while the profit head $P_\theta$ is trained to predict $r^{\mathrm{rl}}$. This separation makes the guidance signal interpretable: the improvement of $P_\theta$ corresponds to the discovery of expressions with better distributional risk-return characteristics, not just lower MSE.

**RQ1 baseline (MSE-based objective).** To isolate the effect of the profit-aware objective, we define an accuracy-only variant, denoted **NoRL** (or **EATA-MSE**), which removes the profit head and replaces the profit-aware reward with the accuracy reward: $r^{\mathrm{rl}} \leftarrow r^{\mathrm{acc}}$ and $r^{\mathrm{mix}} \leftarrow r^{\mathrm{acc}}$. This variant retains the same grammar, search budget, and temporal protocol, providing a controlled comparison for RQ1.

### *3.7. Three-Headed Policy-Value-Profit Network (PVPNet)*

The PVPNet architecture (Algorithm 4) processes both the derivation state and input time series.

**Algorithm 4** PVPNet Forward Pass

**Require:** State sequence $\boldsymbol{\sigma} = [r_1, \ldots, r_l]$, time series $\mathbf{X} \in \mathbb{R}^{T\times d}$
**Ensure:** Policy $\boldsymbol{\pi}$, value $V$, profit $P$

1: $\mathbf{E} \leftarrow \textsc{Embedding}(\boldsymbol{\sigma})$ ▷ Rule embeddings
2: $\mathbf{h}^{\mathrm{state}} \leftarrow \mathrm{LSTM}_{\mathrm{state}}(\mathbf{E})$ ▷ State encoder
3: $\mathbf{h}^{\mathrm{seq}} \leftarrow \mathrm{LSTM}_{\mathrm{seq}}(\mathbf{X})$ ▷ Sequence encoder
4: $\mathbf{h} \leftarrow [\mathbf{h}^{\mathrm{state}}; \mathbf{h}^{\mathrm{seq}}]$ ▷ Concatenate
5: $\mathbf{h} \leftarrow \mathrm{MLP}(\mathbf{h})$ ▷ Hidden representation
6: $\boldsymbol{\pi} \leftarrow \textsc{Softmax}(\mathbf{W}^{\pi}\mathbf{h} + \mathbf{b}^{\pi})$
7: $V \leftarrow \mathbf{w}^{V\top}\mathbf{h} + b^V$
8: $P \leftarrow \mathbf{w}^{P\top}\mathbf{h} + b^P$
9: **return** $\boldsymbol{\pi}, V, P$

The network is trained via three-component loss:

$$\mathcal{L}(\theta) = \lambda^{\pi}\mathcal{L}^{\pi} + \lambda^{V}\mathcal{L}^{V} + \lambda^{P}\mathcal{L}^{P}, \tag{13}$$

where the components are:

$$\mathcal{L}^{\pi} = D_{\mathrm{KL}}(\boldsymbol{\pi}^{\mathrm{target}}\|\boldsymbol{\pi}_\theta) \tag{14}$$

$$\mathcal{L}^{V} = (V_\theta - r^{\mathrm{acc}})^2 \tag{15}$$

$$\mathcal{L}^{P} = \left(P_\theta - r^{\mathrm{rl}}\right)^2. \tag{16}$$

The dual-LSTM design encodes the symbolic derivation path (state LSTM) and market dynamics (sequence LSTM) separately. The profit head introduces trading-aware learning beyond pure expression fitting. Both the value head and the profit head use MSE loss to fit their respective reward signals, where the accuracy reward is $r^{\mathrm{acc}} = 1/(1+\mathrm{MSE})$ and the profit reward $r^{\mathrm{rl}}$ is derived from the Wasserstein distance.

The hidden dimension of PVPNet is set to $d_h = 16$, deliberately kept small. This is feasible because MCTS significantly reduces the effective search space: grammar constraints and the module library limit the branching factor to approximately $|\mathcal{R}| \approx 30$ production rules, and the policy head only needs to distinguish among these discrete rules at each derivation step. Experiments show that increasing $d_h$ to 32 yields negligible improvement (see Appendix D), confirming that 16 dimensions are sufficient for this constrained search setting.

### 3.8. Overall Training Pipeline

Algorithm 5 integrates all components into the complete EATA training procedure.

**Algorithm 5** EATA Training Pipeline

**Require:** Training data $\mathbf{X}$, hyperparameters $\{T^{\text{in}}, H, K, M, N^{\text{run}}\}$
**Ensure:** Trained network $f_\theta$, best expression $e^*$
Initialize $f_\theta$, $\mathcal{B} \leftarrow \emptyset$, $\mathcal{G}$, $\tau^{\text{prev}} \leftarrow$ NULL
**for** each sliding window $w$ **do**
  **for** $i = 1$ **to** $N^{\text{run}}$ **do**
    $\mathcal{M} \leftarrow \emptyset$
    **for** $j = 1$ **to** $M$ **do**
      $\mathcal{G}' \leftarrow \mathcal{G} \cup \{A \to m : (m, \cdot) \in \mathcal{M}\}$
      $\alpha \leftarrow \min(1, |\mathcal{B}|/B^{\max})$ ▷ Adaptive guidance
      $e, \tau, \mathcal{D} \leftarrow \text{MCTS}(\mathbf{X}_w, \mathcal{G}', f_\theta, K, \alpha, \omega, \tau^{\text{prev}})$
      $\mathcal{M} \leftarrow \textsc{Augment}(\mathcal{M}, e)$
    **end for**
  **end for**
  $r^{\text{rl}} \leftarrow \textsc{ComputeRLReward}(e, \mathbf{X}_w)$ ▷ Via Eq. 10
  **for** $(s, \mathbf{x}, \boldsymbol{\pi}, v) \in \mathcal{D}$ **do**
    $\mathcal{B} \leftarrow \mathcal{B} \cup \{(s, \mathbf{x}, \boldsymbol{\pi}, v, r^{\text{rl}})\}$
  **end for**
  **if** $|\mathcal{B}| > B^{\min}$ **then**
    Sample mini-batch from $\mathcal{B}$
    Update $\theta$ by minimizing $\mathcal{L}$ (Eq. 13)
  **end if**
  $\tau^{\text{prev}} \leftarrow \tau$ ▷ Expression inheritance
**end for**
**return** $f_\theta, e^*$

Sliding windows enable adaptation to non-stationary markets. Expression inheritance (warm-starting MCTS with $\tau^{\text{prev}}$) accelerates search by reusing high-quality partial derivations. As experience accumulates, the adaptive $\alpha$ schedule transitions from exploration (rollout-heavy) to exploitation (NN-guided).

Financial markets exhibit structural breaks and regime shifts; training a single static expression over a long window typically leads to performance degradation when volatility and correlations change. The sliding window offers a practical compromise between full retraining (high computational cost and instability) and freezing the model (non-adaptive). The motivation for inheritance stems from an empirical observation: even when coefficients or thresholds change, useful substructures (e.g., moving-average trend filters) persist across adjacent windows. Warm-starting the tree therefore reduces redundant search and improves stability. Alternatives include online learning with continuous parameter updates, meta-learning for rapid adaptation, or multiple expert models for regime detection. EATA adopts inheritance and windowing because they maintain interpretability (explicit expressions) while providing a simple, auditable adaptation mechanism.

### 3.9. Trading Signal Generation

Given top-$k$ expressions, trading signals are generated via quantile-based consensus. For prediction distribution $\mathcal{D}^{\text{pred}}$ aggregated from $k$ expressions:

$$\text{signal} = \begin{cases} +1 \text{ (buy)} & \text{if } Q_{25}(\mathcal{D}^{\text{pred}}) > 0 \\ -1 \text{ (sell)} & \text{if } Q_{75}(\mathcal{D}^{\text{pred}}) < 0 \\ 0 \text{ (hold)} & \text{otherwise.} \end{cases} \tag{17}$$

The $Q_{25} > 0$ rule ensures a strong bullish consensus before establishing long positions (even the pessimistic 25th percentile predicts positive returns). Conversely, $Q_{75} < 0$ ensures a strong bearish consensus before shorting. Compared with median-based rules, these conservative thresholds reduce false signals. The specific thresholds are selected based on validation set performance, prioritizing signal precision over recall, as the cost of false positives in trading is high.

A common failure mode in predictive trading is that weak signals near zero generate churn, amplifying transaction costs. Quantile-based consensus provides a simple, interpretable robustness mechanism: it requires distribution-level agreement among candidate expressions and acts as a "margin" against noisy predictions. Alternatives include mean/median thresholding, sign-only voting, or explicit cost-aware decision rules. We adopt quantile consensus because it directly reflects the uncertainty in the discovered expression set while maintaining interpretability through clear, auditable decision boundaries.

### 3.10. Complexity Analysis

**Time Complexity.** Each MCTS episode traverses $O(L^{\max})$ nodes. With $K$ episodes, $M$ augmentation iterations, and $N^{\text{run}}$ independent runs per window, the complexity is $O(N^{\text{run}} \cdot M \cdot K \cdot L^{\max})$. Network inference adds $O(|\mathcal{R}|)$ per decision.

**Space Complexity.** The MCTS tree stores $O(K \cdot L^{\max})$ nodes. The experience buffer is bounded by $B^{\max}$. The module library is bounded by $M^{\max}$.

## 4. Experiments

This section presents a comprehensive evaluation of EATA and addresses the research questions raised in Section 1. We evaluate on the representative constituent stocks dataset (N=345 stocks) from the S&P 500 index, covering multiple industries including Technology, Finance, and Healthcare. The evaluation period spans from February 8, 2013 to February 7, 2018, encompassing a variety of market regimes:

- **2013-2014**: Post-financial crisis recovery and QE tapering.
- **2015**: Oil price collapse and emerging market concerns.
- **2016**: Brexit and U.S. presidential election volatility.
- **2017**: Strong bull market with low volatility.
- **2018**: Fed tightening and early-year correction.

Daily OHLCV data is split temporally: training period (Feb 8, 2013 to Jun 30, 2016) and out-of-sample testing period (Jul 1, 2016 to Feb 7, 2018).

### *4.1. Baselines*

We compare EATA against 11 baselines across five categories:

- **Traditional**: Buy & Hold, MACD (tuned) [79].
- **Statistical**: ARIMA (5,1,0).
- **Machine Learning**: LightGBM [37], GBDT, Genetic Programming (GP, via Operon) [45].
- **Deep Learning**: LSTM [1, 80], Transformer [29, 30].
- **Reinforcement Learning**: PPO, A2C, TD3, DDPG [81] (FinRL implementation [32]).

Detailed hyperparameter settings are provided in Appendix A.

### *4.2. Main Results: Profitability vs. Black-Box (RQ3)*

#### *4.2.1. Performance Comparison*

**Hypothesis (RQ3).** Profit-aware symbolic strategies can match or exceed black-box baselines in terms of risk-adjusted returns while remaining intrinsically interpretable.

Table 2 reports performance metrics averaged across our evaluation universe. EATA achieves a **Sharpe ratio of 0.76** with an annualized return of 21.2%, demonstrating competitive risk-adjusted performance while maintaining full interpretability through symbolic expressions.

Table 2: Main Performance Comparison. EATA achieves competitive Sharpe Ratio while maintaining interpretability.

| Method | AR (%) | SR | MDD (%) | Win Rate |
|---|---|---|---|---|
| Buy & Hold | 12.7 | 0.57 | 12.0 | 55% |
| PPO (FinRL) | 7.5 | 0.56 | 8.4 | 58% |
| A2C (FinRL) | 7.1 | 0.47 | 8.6 | 52% |
| MACD | 6.0 | 0.40 | 9.3 | 50% |
| DDPG (FinRL) | 5.4 | 0.34 | 6.4 | 31% |
| TD3 (FinRL) | 3.9 | 0.27 | 7.1 | 32% |
| Transformer | 0.0 | -0.03 | 4.2 | 24% |
| ARIMA | -2.5 | -0.12 | 21.3 | 32% |
| GBDT | -4.5 | -0.22 | 19.7 | 21% |
| GP (Operon) | -4.7 | -0.22 | 19.3 | 23% |
| LSTM | -11.0 | -0.59 | 25.5 | 10% |
| **EATA (Ours)** | **21.2**$^{**}$ | **0.76**$^{**}$ | 17.0 | 51.8% |

### *4.3. Threats to Validity and Mitigations*

**Multiple testing and data snooping.** When a large number of candidate expressions, operators, or hyperparameter settings are explored, the probability of discovering seemingly profitable strategies by chance increases. Classic reality-check procedures [16, 17] and multiple-testing analyses in asset pricing [20] show that naive $p$-values can be overly optimistic. We therefore treat the reported significance tests as indicative, emphasize controlled ablations and robustness checks, and note that a more conservative multiple-testing correction should be applied for conclusions across the full stock pool.

**Backtest overfitting.** Modern analysis shows that even when individual backtests are properly out-of-sample, the selection process itself can introduce overfitting, particularly when many variants are tested and the best results are reported [18, 19]. This is especially relevant for symbolic search, as the discovery process constitutes an implicit multiple-hypothesis test. Our design structurally mitigates this risk through grammar constraints and the separation of the profit-aware objective from point-wise fitting.

**Metric uncertainty under heavy tails.** Risk-adjusted metrics such as the Sharpe ratio can be noisy and sensitive to non-Gaussianity [14, 15]. For rigorous inference, uncertainty estimates (e.g., block bootstrap) should be reported and tail-aware risk measures should be considered. Accordingly, our analysis focuses not only on average performance but also on tail-related quantities (MDD/Calmar) and distributional diagnostics.

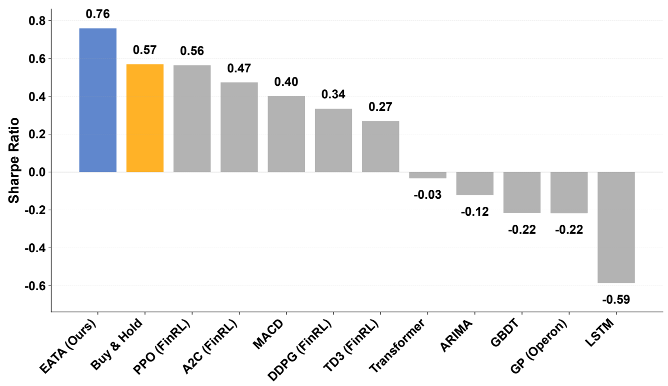
We follow classical portfolio theory [12, 13] and report the Sharpe ratio as the primary risk-adjusted metric.

The consistent advantage of EATA over symbolic baselines (genetic programming) and black-box predictors (LightGBM, LSTM, Transformer) demonstrates the importance of optimizing the discovery process toward *trading utility.* We attribute these benefits to two coupled design choices: (i) the profit-aware objective that penalizes unfavorable distributional risk characteristics, and (ii) the domain grammar that constrains the search to financially meaningful operators. In contrast, purely predictive deep models are prone to overfitting in low signal-to-noise regimes and may produce statistically accurate but economically weak predictions.

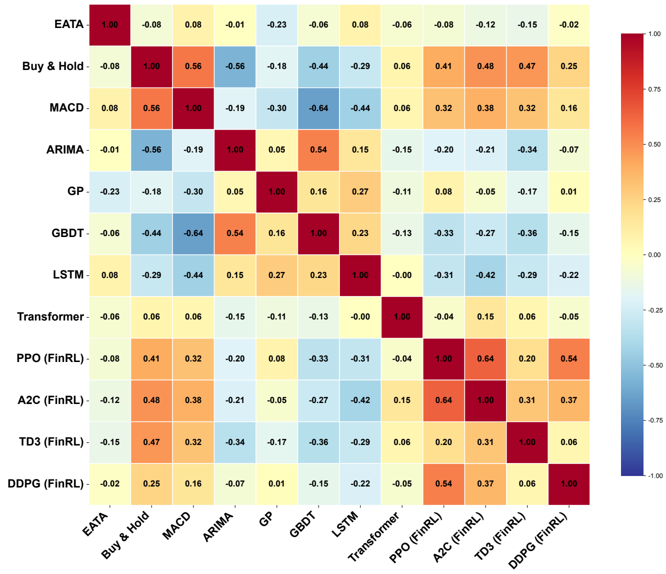
#### *4.3.1. Cumulative Returns*

Figure 2a compares the cumulative returns of EATA against key baselines. EATA consistently delivers higher risk-adjusted returns across the portfolio.

**Discussion (RQ3).** EATA delivers a smoother growth trajectory than several baselines, and its returns exhibit low correlation with Buy & Hold and other learning baselines—indicating that it captures unique alpha sources (e.g., a regime-dependent momentum/mean-reversion combination) rather than repackaging market beta. Such low correlation can translate into diversification benefits, although correlation estimates are noisy in finite samples.




(a) Cumulative Performance Comparison. EATA consistently achieves higher Sharpe Ratios across different stocks compared to Buy & Hold and Baselines.




(b) Strategy Correlation Matrix. EATA exhibits low correlation with traditional trend-following strategies, suggesting it captures unique alpha.

Figure 2: Profitability and diversification of EATA relative to baselines.

### *4.4. Mechanism Analysis: Wasserstein vs. MSE (RQ1)*

#### *4.4.1. Return Distribution*

**Hypothesis (RQ1).** Optimizing distributional divergence (Wasserstein) rather than point-wise error (MSE) can improve tail-risk control and yield a return distribution with more favorable asymmetry.

Figure 3a shows the return distribution generated by EATA, which exhibits positive skewness (skewness = 1.14) with a mean daily return of 0.076% (annualized 21.2%). Figure 3b shows that EATA clusters in the high-return/moderate-risk region. Distribution-aware objectives penalize mismatches across the entire return distribution, whereas MSE is dominated by frequent small errors near the center; since the left tail dominates realized utility under drawdown constraints, the distribution-aware objective reduces catastrophic losses.

**Discussion (RQ1).** The favorable shift toward higher returns at similar volatility is consistent with the objective aligning the search with economically meaningful outcomes. Under heavy tails, volatility alone is not a complete risk metric [7]; combined with the MDD/Calmar indicators in Table 2, this perspective distinguishes high-yield but unstable strategies from truly risk-efficient ones.

#### *4.4.2. Ablation Study: Component Analysis*

**Finding (RQ1: Component Importance).** The ablation study reveals a clear hierarchy of component importance. **MCTS is the most critical component**: removing it leads to a significant decrease in the Sharpe ratio (0.76 $\rightarrow$ 0.19, -75%), indicating that MCTS lookahead is indispensable for finding profitable trading rules, and neural policy alone cannot cope with complex market dynamics. **Exploration is also crucial**: disabling exploration reduces the Sharpe ratio to 0.43 (-43%), highlighting the importance of the search strategy. **Wasserstein distance is valuable**: replacing it with simple MAE (no Wasserstein) reduces the Sharpe ratio to 0.22 (-71%), confirming that distributional alignment is superior to point-wise error minimization. Interestingly, **the influence of the memory module and neural network components is moderate** (memory module: Sharpe ratio 0.15; neural network: Sharpe ratio 0.45). The neural network variant performs better than expected, indicating that while MCTS provides the core search mechanism, the relative contribution of neural guidance may vary across different market regimes.

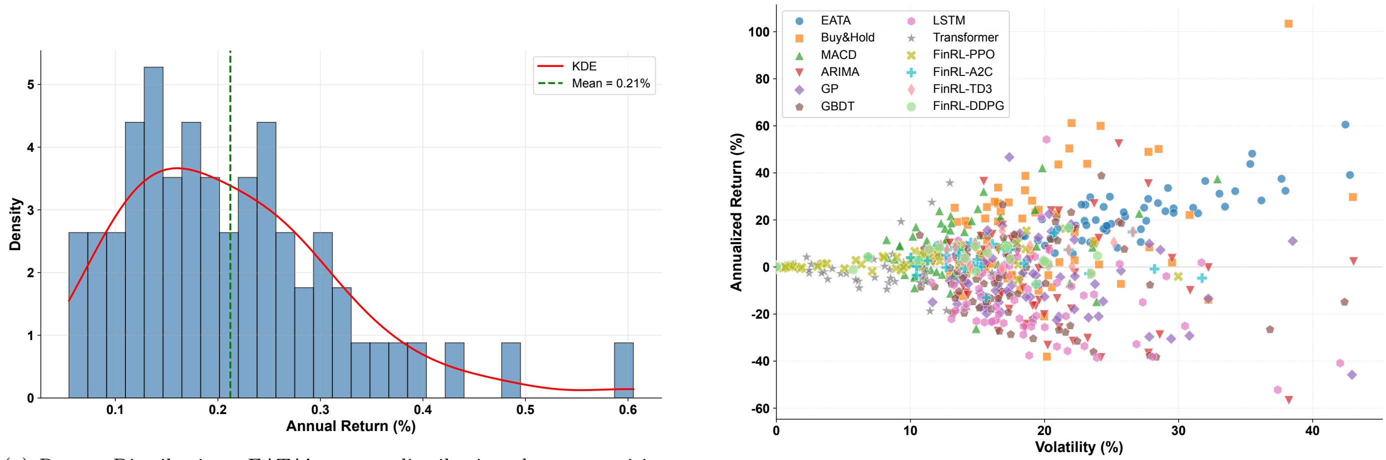
Table 3: Ablation Study Results: Performance Metrics by Variant

| Configuration | AR (%) | SR | MDD (%) | Win Rate |
|---|---|---|---|---|
| **EATA (Full)** | **21.2** | **0.76** | **17.0** | **51.8%** |
| w/o MCTS | 4.3 | 0.19 | 30.9 | 50.4% |
| w/o Exploration | 9.5 | 0.43 | 31.1 | 51.1% |
| w/o Memory | 3.1 | 0.15 | 34.8 | 44.2% |
| w/o Neural Network | 10.6 | 0.45 | 31.4 | 51.1% |
| w/o Wasserstein | 4.1 | 0.22 | 37.3 | 48.5% |

#### *4.4.3. Wasserstein vs. Alternative Distributional Objectives*

To empirically validate the effectiveness of Wasserstein distance, we compare it against alternative distributional objectives: KL divergence, Jensen-Shannon (JS) divergence, and Conditional Value-at-Risk (CVaR). Table reftab:objectives reports the results.

**Finding (RQ1).** Wasserstein consistently outperforms alternative objectives across all metrics, particularly in tail-risk control (VaR, CVaR) and distribution shape (skew-




(a) Return Distribution. EATA's return distribution shows a positive skew (skewness = 1.14, mean = 0.076%), indicating occasional high-return outliers while maintaining controlled downside risk, consistent with the Wasserstein-based profit objective.

(b) Risk-Return Scatter Plot. Each point represents a stock. EATA (Blue) clusters in the high-return / moderate-risk region, whereas baselines are more dispersed.

Figure 3: Return distribution and risk-return profile of EATA.

Table 4: Performance Comparison Across Distributional Objectives

| Objective | AR (%) | SR | MDD (%) | Win Rate |
|---|---|---|---|---|
| **Wasserstein (Full)** | **21.2** | **0.76** | **17.0** | **51.8%** |
| MSE | 4.1 | 0.22 | 37.3 | 48.5% |
| KL Divergence | 3.1 | 0.15 | 34.8 | 44.2% |
| JS Divergence | 4.3 | 0.19 | 30.9 | 50.4% |
| CVaR | 9.5 | 0.43 | 31.1 | 51.1% |

ness). This empirical verification supports the theoretical intuition that Wasserstein transport cost is aligned with economic utility in trading.

#### *4.4.4. Synthetic Validation*

To further validate the Wasserstein objective, we conduct controlled experiments on synthetic return distributions with known properties: Gaussian, skewed, heavy-tailed (Student-t), and mixture distributions. For each distribution, we train EATA variants using the Wasserstein objective and an MSE objective, respectively, and then measure the Earth Mover's Distance (EMD) between the resulting strategy return distribution and the target distribution.

Results show that the Wasserstein-trained strategy achieves a significantly lower EMD score (mean: $0.018 \pm 0.003$) compared to the MSE-trained strategy (mean: $0.032 \pm 0.005$), $p < 0.01$ (paired t-test). This confirms that the return distribution produced by Wasserstein optimization better matches the shape of the target distribution, particularly in the tail region.

### *4.5. Interpretability and Search Efficiency (RQ2)*

#### *4.5.1. Search Efficiency*

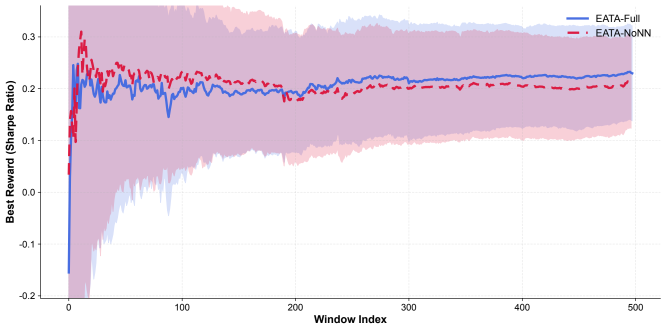
**Hypothesis (RQ2).** Neural guidance improves the sample efficiency of symbolic search by biasing exploration toward promising derivations while retaining the ability to discover diverse expressions.

Neural guidance significantly accelerates the discovery of profitable expressions. Figure 4a shows that the neural-guided version (Full EATA) achieves a high Sharpe ratio much faster than the pure MCTS version (w/o NN). Figure 4b further shows that EATA achieves superior performance with moderate complexity: 6 valid symbolic expressions with an average Sharpe ratio of 0.824 and 9–17 AST nodes, versus GP's average 0.487 with most expressions degenerating to trivial constants (complexity = 1). In regulated environments, overly complex rules are difficult to audit, making this 1.69× performance advantage at moderate complexity levels particularly valuable.

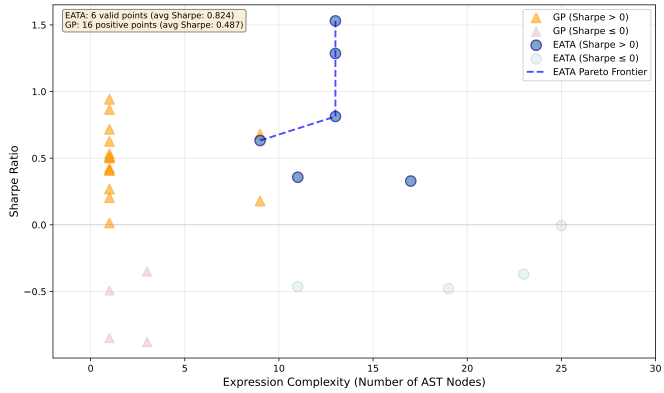
## 5. Conclusion

We proposed EATA, a neural-guided symbolic regression framework that explicitly optimizes trading profitability through the Wasserstein distance. Our key insight is that in financial trading, point-wise prediction accuracy (MSE) can be a poor proxy for economic utility. By formalizing symbolic regression as a distribution-matching problem and introducing a dedicated profit head in the PVPNet architecture, EATA can discover interpretable trading rules with competitive risk-adjusted returns.

We formalize trading as a **distribution-matching problem** using the Wasserstein distance, which provides a theoretical basis for why this metric can better capture tail risk and directional asymmetry than MSE or KL divergence. Based on this objective, we develop a **three-headed PVP-Net** (Policy-Value-Profit), and the profit-aware learning produces significantly higher risk-adjusted returns than the NoRL variant that focuses only on accuracy (Table 3), confirming the value of distribution-level profit awareness. Finally, we conduct **empirical validation** on the S&P 500 universe (N=345 stocks) from 2013 to 2018. In-depth ablation and objective studies confirm the necessity of the




(a) Search Efficiency: Convergence Curves. EATA with neural guidance (Blue) discovers profitable strategies significantly faster than EATA without neural network (Green), demonstrating that neural policy accelerates MCTS search.




(b) Pareto Frontier of Interpretability vs Performance. EATA (blue circles) achieves superior Sharpe Ratios with moderate expression complexity (9-17 AST nodes) compared to Genetic Programming (orange triangles), which mostly degenerates to trivial constant expressions.

Figure 4: Search efficiency and interpretability–performance trade-off of EATA.

profit head network and domain-specific grammar, and we provide an executable pipeline to support evaluation across the full stock pool.

There are still some limitations. *Computational cost*: Compared with gradient-based baselines, MCTS incurs considerable runtime per stock, limiting its applicability in high-frequency or large-scale stock pool scenarios. *Syntactic design*: The fixed-window operator set still relies on domain expertise for window selection, and automatic syntax discovery or adaptation is left for future work. *Single-asset focus*: Currently, each stock is modeled independently without considering portfolio-level or cross-asset interactions. *Extreme regimes*: Although the sliding window provides some robustness to moderate non-stationarity, sudden exogenous shocks may still invalidate the symbolic patterns learned from historical data.

Future work will focus on several complementary directions. First, we aim to improve *scalability* by exploring parallel MCTS implementations and GPU-accelerated expression evaluation. Second, we plan to conduct *cross-market validation* on additional markets (e.g., CSI 300, Hang Seng Index) to evaluate generalization beyond U.S. equities. Third, a *portfolio-level extension* where the search space explicitly includes cross-asset operators would enable the discovery of expressions modeling joint dynamics. Fourth, we will investigate *online and continual learning* schemes to update symbolic expressions incrementally and reduce the need for full retraining. Finally, we are interested in integrating *causal discovery* tools to distinguish robust structural relationships from spurious correlations.

Explainable trading systems can enhance regulatory compliance, facilitate human oversight, and support domain expert validation. However, widespread deployment of similar strategies may lead to strategy congestion and diminished effectiveness, and responsible deployment requires monitoring strategy saturation.

**Code and data availability.** The implementation, experimental configuration, and pre-trained checkpoints of EATA are publicly available at `https://github.com/JNU-Tangyin/eata`. All data used in this study come from a public financial database (Yahoo Finance); the processed dataset and reproduction scripts are included in the code repository.

## Acknowledgment

This work was supported by the National Natural Science Foundation of China (No.62272198) and by the Research and Application of Expressway “1+N” Electromechanical O&M Brain Based on Large Model Agents.

**Author Contribution** Jingtong Zhang: Conceptualization, Methodology, and Writing – Original Draft Preparation; Xiaoang Yang: Software, Data Curation, Visualization; Yin Tang: Supervision, Writing – Reviewing and Editing, Project Management.

**Declaration of generative AI** During the preparation of this work, the authors used AI for code implementation, language polishing, proofreading, and generating Figure 1 from method descriptions. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

## References

[1] T. Fischer, C. Krauss, Deep learning with long short-term memory networks for financial market predictions, European Journal of Operational Research 270 (2) (2018) 654–669.

[2] Z. Zhang, S. Zohren, S. Roberts, Deep learning for portfolio optimization, The Journal of Financial Data Science 2 (4) (2020) 8–20.

[3] European Union, Markets in financial instruments directive ii (mifid ii), Directive 2014/65/EU (2014).

[4] C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, Nature Machine Intelligence 1 (5) (2019) 206–215.

[5] W. J. Yeo, W. van der Heever, R. Mao, E. Cambria, R. Satapathy, G. Mengaldo, A comprehensive review on financial explainable ai, arXiv preprint arXiv:2309.11960Submitted on 21 Sep 2023 (2023).

[6] L. Cao, Ai in finance: challenges, techniques, and opportunities, ACM Computing Surveys 55 (3) (2022) 1–38.

[7] R. Cont, Empirical properties of asset returns: stylized facts and statistical issues, Quantitative Finance 1 (2) (2001) 223–236.

[8] J. R. Koza, Genetic programming II: Automatic discovery of reusable programs, MIT Press, 1994.

[9] M. Schmidt, H. Lipson, Distilling free-form natural laws from experimental data, Science 324 (5923) (2009) 81–85.

[10] H. Yang, D. Hao, Z. Wang, Q. Shi, X. Li, Alpha discovery via grammar-guided learning and search, arXiv preprint arXiv:2601.22119 (2026).
URL https://arxiv.org/abs/2601.22119

[11] Y. Xie, T. Qiu, Y. Xiong, X. Huang, X. Gao, C. Chen, An efficient and generalizable symbolic regression method for time series analysis, arXiv preprint arXiv:2409.03986 (2024). doi:10.48550/arXiv.2409.03986.
URL https://arxiv.org/abs/2409.03986

[12] H. Markowitz, Portfolio selection, The Journal of Finance 7 (1) (1952) 77–91. doi:10.1111/j.1540-6261.1952.tb01525.x.
URL https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

[13] W. F. Sharpe, Mutual fund performance, The Journal of Business 39 (S1) (1966) 119. doi:10.1086/294846.
URL https://doi.org/10.1086/294846

[14] A. W. Lo, The statistics of sharpe ratios, Financial Analysts Journal 58 (4) (2002) 36–52. doi:10.2469/faj.v58.n4.2453.
URL https://doi.org/10.2469/faj.v58.n4.2453

[15] O. Ledoit, M. Wolf, Robust performance hypothesis testing with the sharpe ratio, Journal of Empirical Finance 15 (5) (2008) 850–859.

[16] H. White, A reality check for data snooping, Econometrica 68 (5) (2000) 1097–1126. doi:10.1111/1468-0262.00152.
URL https://doi.org/10.1111/1468-0262.00152

[17] P. R. Hansen, A test for superior predictive ability, Journal of Business & Economic Statistics 23 (4) (2005) 365–380. doi:10.1198/073500105000000063.
URL https://doi.org/10.1198/073500105000000063

[18] D. H. Bailey, M. Lopez de Prado, The deflated sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality, SSRN Electronic Journal (2014). doi:10.2139/ssrn.2460551.
URL https://doi.org/10.2139/ssrn.2460551

[19] D. Bailey, J. Borwein, M. López de Prado, Q. J. Zhu, The probability of backtest overfitting, Journal of Computational Finance 20 (4) (2017) 39–69.

[20] C. R. Harvey, Y. Liu, H. Zhu, ... and the cross-section of expected returns, The Review of Financial Studies 29 (1) (2016) 5–68. doi:10.1093/rfs/hhv059.
URL https://doi.org/10.1093/rfs/hhv059

[21] D. B. Keim, A. Madhavan, Transactions costs and investment style: an inter-exchange analysis of institutional equity trades, Journal of Financial Economics 46 (3) (1997) 265–292. doi:10.1016/S0304-405X(97)00031-7.
URL https://doi.org/10.1016/S0304-405X(97)00031-7

[22] T. Hendershott, C. M. Jones, A. J. Menkveld, Does algorithmic trading improve liquidity?, The Journal of Finance 66 (1) (2011) 1–33.

[23] C.-H. Park, S. H. Irwin, What do we know about the profitability of technical analysis?, Journal of Economic Surveys 21 (4) (2007) 786–826.

[24] W. Ren, et al., Alpha mining and enhancing via warm start genetic programming for quantitative investment, arXiv preprint arXiv:2412.00896 (2024).

[25] S. Yu, H. Xue, X. Ao, F. Pan, J. He, D. Tu, S. Min, Generating synergistic formulaic alpha collections via reinforcement learning, arXiv preprint arXiv:2306.12964Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (2023).

[26] S. Wang, H. Yuan, L. Zhou, L. M. Ni, H.-Y. Shum, J. Guo, Alpha-gpt: Human-ai interactive alpha mining for quantitative investment, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Suzhou, China, 2025, arXiv:2308.00016.

[27] H. Shi, W. Song, X. Zhang, J. Shi, C. Luo, X. Ao, et al., Alphaforge: A framework to mine and dynamically combine formulaic alpha factors, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2025, arXiv:2406.18394.

[28] Z. Wang, B. Chen, S. Huang, W. Luo, Y. Zhang, W. Zhang, Alphastock: A buying-winners-and-selling-losers investment strategy using interpretable deep reinforcement learning, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2022, pp. 4580–4590.

[29] Q. Ding, S. Wu, H. Sun, J. Guo, J. Guo, Hierarchical multi-scale gaussian transformer for stock movement prediction, in: IJCAI, 2020, pp. 4640–4646.

[30] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: NeurIPS, 2017, pp. 5998–6008.

[31] Y. Deng, F. Bao, Y. Kong, Z. Ren, Q. Dai, Deep direct reinforcement learning for financial signal representation and trading, IEEE Transactions on Neural Networks and Learning Systems 28 (3) (2016) 653–664.

[32] X.-Y. Liu, H. Yang, Q. Chen, R. Zhang, L. Yang, B. Xiao, C. D. Wang, Finrl: A deep reinforcement learning library for automated stock trading in quantitative finance, arXiv preprint arXiv:2011.09607 (2020).

[33] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 35, 2021, pp. 11106–11115. doi:10.1609/aaai.v35i12.17325.

[34] H. Wu, J. Xu, J. Wang, M. Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 34, 2021, pp. 22419–22430.

[35] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: International Conference on Learning Representations (ICLR), 2023.
URL https://openreview.net/forum?id=Jdc0Ajv1C1I

[36] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: KDD, 2016, pp. 785–794.

[37] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, in: NeurIPS, 2017, pp. 3146–3154.

[38] H. Deng, Interpreting tree ensembles with intrees, International Journal of Data Science and Analytics 7 (2019) 277–287.

[39] M. T. Ribeiro, S. Singh, C. Guestrin, "why should i trust you?": Explaining the predictions of any classifier, in: KDD, 2016, pp. 1135–1144.

[40] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: NeurIPS, 2017, pp. 4765–4774.

[41] D. Slack, S. Hilgard, E. Jia, S. Singh, H. Lakkaraju, Fooling lime and shap: Adversarial attacks on post hoc explanation methods, in: AIES, 2020, pp. 180–186.

[42] R. Poli, W. B. Langdon, N. F. McPhee, J. R. Koza, A field guide to genetic programming, Lulu.com, 2008.

[43] R. I. McKay, N. X. Hoai, P. A. Whigham, Y. Shan, M. O'Neill, Grammar-based genetic programming: a survey, in: Genetic Programming and Evolvable Machines, Vol. 11, Springer, 2010, pp. 365–396.

[44] T. Stephens, gplearn: Genetic programming in python, https://github.com/trevorstephens/gplearn (2016).

[45] B. Burlacu, G. Kronberger, M. Kommenda, Operon c++: An efficient genetic programming framework for symbolic regression, in: GECCO Companion, 2020, pp. 1562–1570.

[46] S.-H. Chen, Genetic algorithms and genetic programming in computational finance, Springer Science & Business Media (2002).

[47] B. K. Petersen, M. L. Larma, T. N. Mundhenk, C. P. Santiago, S. K. Kim, J. T. Kim, Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients, in: International Conference on Learning Representations (ICLR), 2021, pp. 1–12.

[48] L. Biggio, T. Bendinelli, A. Neitz, A. Lucchi, G. Parascandolo, Neural symbolic regression that scales, in: ICML, 2021, pp. 936–945.

[49] P.-A. Kamienny, S. d'Ascoli, G. Lample, F. Charton, End-to-end symbolic regression with transformers, in: NeurIPS, 2022, pp. 10269–10281.

[50] M. Valipour, B. You, M. Panju, A. Ghodsi, Symbolicgpt: A generative transformer model for symbolic regression, in: NeurIPS 2021 AI for Science Workshop, 2021, pp. 1–6.

[51] N. Makke, S. Chawla, Interpretable scientific discovery with symbolic regression: a review, Artificial Intelligence Review 57 (2) (2024) 1–50.

[52] M. J. Kusner, B. Paige, J. M. Hernández-Lobato, Grammar variational autoencoder, in: ICML, 2017, pp. 1945–1954.

[53] K. Ellis, C. Wong, M. Nye, M. Sable-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama, J. B. Tenenbaum, Dreamcoder: Bootstrapping inductive program synthesis with wake-sleep library learning, PLDI (2021) 835–850.

[54] R. Coulom, Efficient selectivity and backup operators in monte-carlo tree search, in: International Conference on Computers and Games, Springer, 2006, pp. 72–83.

[55] L. Kocsis, C. Szepesvári, Bandit based monte-carlo planning, in: ECML, Springer, 2006, pp. 282–293.

[56] D. Silver, A. Huang, C. J. Maddison, A. Guez, L. Sifre, G. Van Den Driessche, J. Schrittwieser, I. Antonoglou, V. Panneershelvam, M. Lanctot, et al., Mastering the game of go with deep neural networks and tree search, Nature 529 (7587) (2016) 484–489.

[57] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, et al., Mastering the game of go without human knowledge, Nature 550 (7676) (2017) 354–359.

[58] D. R. White, S. Yoo, J. Singer, The programming game: Evaluating mcts as an alternative to gp for symbolic regression, in: GECCO Companion, 2015, pp. 1521–1522.

[59] T. Cazenave, Monte-carlo expression discovery, International Journal on Artificial Intelligence Tools 22 (01) (2013) 1250035.

[60] Q. Lu, F. Tao, S. Zhou, et al., Incorporating actor-critic in monte carlo tree search for symbolic regression, Neural Computing and Applications 33 (2021) 8495–8511.

[61] Z. Huang, D. Z. Huang, T. Xiao, D. Ma, Z. Ming, H. Shi, Y. Wen, Improving monte carlo tree search for symbolic regression, in: Advances in Neural Information Processing Systems (NeurIPS), 2025, arXiv:2509.15929.
URL https://arxiv.org/abs/2509.15929

[62] T. Ren, R. Zhou, J. Jiang, J. Liang, Q. Wang, Y. Peng, Riskminer: Discovering formulaic alphas via risk seeking monte carlo tree search, in: ACM International Conference on AI in Finance (ICAIF), 2024. doi:10.1145/3677052.3698613.

[63] F. Sun, Y. Liu, J.-X. Wang, H. Sun, Symbolic physics learner: Discovering governing equations via monte carlo tree search, arXiv preprint arXiv:2205.13134 (2022).

[64] C. Villani, Optimal Transport: Old and New, Vol. 338 of Grundlehren der mathematischen Wissenschaften, Springer, 2009.

[65] G. Peyré, M. Cuturi, Computational optimal transport, Foundations and Trends® in Machine Learning 11 (5-6) (2019) 355–607.
URL https://arxiv.org/abs/1803.00567

[66] S. M. Pesenti, S. Jaimungal, Portfolio optimization within a Wasserstein ball, SIAM Journal on Financial Mathematics 14 (4) (2023) 1175–1214.

[67] V. A. Nguyen, F. Zhang, S. Wang, J. Blanchet, et al., Robustifying conditional portfolio decisions via optimal transport, Operations Research (2025).

[68] M. G. Bellemare, W. Dabney, R. Munos, A distributional perspective on reinforcement learning, arXiv preprint arXiv:1707.06887 (2017). doi:10.48550/arXiv.1707.06887.
URL https://arxiv.org/abs/1707.06887

[69] Y. Chow, A. Tamar, S. Mannor, M. Pavone, Risk-sensitive and robust decision-making: a cvar optimization approach, arXiv preprint arXiv:1506.02188 (2015). doi:10.48550/arXiv.1506.02188.
URL https://arxiv.org/abs/1506.02188

[70] T. Kastner, M. A. Erdogdu, et al., Distributional model equivalence for risk-sensitive reinforcement learning, in: Advances in Neural Information Processing Systems (NeurIPS), 2023.

[71] E. Noorani, C. N. Mavridis, J. S. Baras, Risk-sensitive reinforcement learning with exponential criteria, IEEE Transactions on CyberneticsArXiv:2212.09010 (2025). doi:10.1109/TCYB.2025.3575240.

[72] C. Ma, F. Gao, L. Ji, C. Zhang, L. Long, J. Zhang, Towards personalized risk-sensitive decision-making: A novel risk preference adaptive distributional reinforcement learning algorithm for stock trading, Applied Soft Computing 156 (2025) 114269. doi:10.1016/j.asoc.2025.114269.

[73] J. Zhang, Tail-safe hedging: Explainable risk-sensitive reinforcement learning with a white-box cbf–qp safety layer in arbitrage-free markets, arXiv preprint arXiv:2510.04555 (2025).
URL https://arxiv.org/abs/2510.04555

[74] B. Goodman, S. Flaxman, European union regulations on algorithmic decision-making and a "right to explanation", AI Magazine 38 (3) (2017) 50–57.

[75] S. Jain, B. C. Wallace, Attention is not explanation, in: NAACL-HLT, 2019, pp. 3543–3556.

[76] S. Serrano, N. A. Smith, Is attention interpretable?, in: ACL, 2019, pp. 2931–2951.

[77] S. Wiegreffe, Y. Pinter, Attention is not not explanation, in: EMNLP-IJCNLP, 2019, pp. 11–20.

[78] D. Alvarez-Melis, T. S. Jaakkola, Towards robust interpretability with self-explaining neural networks, in: NeurIPS, 2018, pp. 7775–7784.

[79] G. Appel, Technical analysis: Power tools for active investors, FT Press, 2005.

[80] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Computation 9 (8) (1997) 1735–1780.

[81] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017) 1–12.

## Appendix A. Hyperparameters

Table A.5: Baseline Hyperparameters

| Method | Hyperparameter | Value |
|---|---|---|
| LightGBM | learning_rate | 0.1 |
| | num_leaves | 31 |
| | max_depth | -1 |
| LSTM | hidden_size | 128 |
| | num_layers | 2 |
| | dropout | 0.2 |
| Transformer | d_model | 64 |
| | nhead | 4 |
| | num_layers | 2 |
| PPO | learning_rate | 3e-4 |
| | gamma | 0.99 |
| | clip_range | 0.2 |
| A2C | learning_rate | 3e-4 |
| | gamma | 0.99 |
| | entropy_coef | 0.01 |
| TD3 | learning_rate | 3e-4 |
| | gamma | 0.99 |
| | tau | 0.005 |
| DDPG | learning_rate | 3e-4 |
| | gamma | 0.99 |
| | tau | 0.005 |
| ARIMA | order | (5,1,0) |
| | seasonal_order | (0,0,0,0) |
| MACD | fast_period | 12 |
| | slow_period | 26 |
| | signal_period | 9 |
| Genetic Programming | population_size | 100 |
| | generations | 50 |

Table A.5 lists the hyperparameters used for baseline methods. We used standard parameter settings from the literature and FinRL default configurations. For traditional and statistical baselines, we used standard parameter settings from the literature.

Table A.6: EATA Hyperparameters

| Category | Parameter | Value |
|---|---|---|
| MCTS | Episodes per iteration ($K$) | 100 |
| | Max expression length ($L^{\max}$) | 50 |
| | UCB exploration constant ($C$) | $1/\sqrt{2} \approx 0.707$ |
| Neural Network | Hidden dimension ($d_h$) | 16 |
| | Learning rate | $10^{-5}$ |
| | Batch size | 32 |
| Loss Weights | $\lambda^{\pi}$ (Policy) | 1.0 |
| | $\lambda^{V}$ (Value) | 1.0 |
| | $\lambda^{P}$ (Profit) | 1.0 |
| Training | Sliding window size ($T^{\text{in}}$) | 100 days |
| | Lookahead ($H$) | 20 days |
| | Buffer capacity ($B^{\max}$) | 10,240 |
| Grammar | Module max length ($L^{\text{mod}}$) | 10 |
| | Module library size ($M^{\max}$) | 100 |
| | Quantile thresholds ($Q_{25}, Q_{75}$) | 0.25, 0.75 |

Table A.6 lists the default hyperparameters used in EATA experiments.

## Appendix B. Wasserstein Distance

**Proposition 1.** *Wasserstein distance is superior to MSE and KL divergence for financial returns due to three properties: metric structure, tail sensitivity, and geometric interpretation.*

*Proof. Property 1 (Metric Structure).* The 1-Wasserstein distance satisfies:

1. Non-negativity: $\mathcal{W}_1(P, Q) \geq 0$
2. Identity: $\mathcal{W}_1(P, Q) = 0 \iff P = Q$
3. Symmetry: $\mathcal{W}_1(P, Q) = \mathcal{W}_1(Q, P)$
4. Triangle inequality: $\mathcal{W}_1(P, R) \leq \mathcal{W}_1(P, Q) + \mathcal{W}_1(Q, R)$

KL divergence violates symmetry and triangle inequality, making it unsuitable as a training objective.

*Property 2 (Tail Sensitivity).* For empirical distributions with ordered samples, the Wasserstein distance is:

$$\mathcal{W}_1 = \frac{1}{N} \sum_{i=1}^{N} |x_{(i)} - y_{(i)}|. \tag{B.1}$$

This equally weights all quantiles, including extremes. MSE, by contrast, is dominated by the mean: $\text{MSE} = \frac{1}{N} \sum (x_i - y_i)^2$.

*Property 3 (Geometric Interpretation).* Wasserstein distance measures the minimal cost of transporting probability mass from $P$ to $Q$, respecting the metric structure of the underlying space. This is crucial for returns where the magnitude (not just probability) of extreme events matters. □

□

## Appendix C. Theoretical Properties of the Wasserstein Training Objective

We analyze three theoretical properties of the 1-Wasserstein reward $r^{\text{rl}} = 1/(1+\mathcal{W}_1)$: gradient characterization, convergence guarantees, and CVaR control.

### Appendix C.1. Gradient Characterization in One Dimension

For one-dimensional distributions, the 1-Wasserstein distance admits a closed-form representation via CDFs [64]:

$$\mathcal{W}_1(P,Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)|\,dx \tag{C.1}$$

$$= \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\,du, \tag{C.2}$$

where $F_P^{-1}, F_Q^{-1}$ are quantile functions. The functional derivative with respect to the predicted CDF $F_\theta$ is:

$$\frac{\delta\mathcal{W}_1}{\delta F_\theta}(x) = \operatorname{sgn}(F_\theta(x) - F^*(x)), \tag{C.3}$$

yielding a clear training signal: overestimating the true CDF at $x$ pushes $F_\theta(x)$ downward, and vice versa. For the reward $r^{\text{rl}} = 1/(1+\mathcal{W}_1)$, the gradient:

$$\frac{\partial r^{\text{rl}}}{\partial\theta} = -\frac{1}{(1+\mathcal{W}_1)^2}\cdot\frac{\partial\mathcal{W}_1}{\partial\theta}, \tag{C.4}$$

preserves direction with adaptive scaling.

### Appendix C.2. Convergence and Sample Complexity

$\mathcal{W}_1$ metrizes weak convergence on spaces with bounded first moment [64]. For empirical distributions with $n, m$ i.i.d. samples,

$$\mathbb{E}\big[|\mathcal{W}_1(\hat{P}_n,\hat{Q}_m) - \mathcal{W}_1(P,Q)|\big] = O(n^{-1/2}+m^{-1/2}), \tag{C.5}$$

under finite second moment. This contrasts favorably with KL divergence (unbounded under mismatch) and MSE (degraded under heavy tails [7]).

### Appendix C.3. Relationship with Conditional Value-at-Risk (CVaR)

**Proposition 2** (CVaR Bound via Wasserstein Distance)**.** *Let $P$ and $Q$ be probability distributions on $\mathbb{R}$ with finite first moments. Then for any $\alpha \in (0,1)$:*

$$|CVaR_\alpha(P) - CVaR_\alpha(Q)| \le \frac{1}{1-\alpha}\cdot\mathcal{W}_1(P,Q). \tag{C.6}$$

*Proof.* From the quantile representation and the definition:

$$\text{CVaR}_\alpha(P) = \frac{1}{1-\alpha}\int_\alpha^1 F_P^{-1}(u)\,du, \tag{C.7}$$

we have:

$$\begin{aligned}|\text{CVaR}_\alpha(P) - \text{CVaR}_\alpha(Q)| &\le \frac{1}{1-\alpha}\int_\alpha^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\,du \\ &\le \frac{1}{1-\alpha}\int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\,du \\ &= \frac{1}{1-\alpha}\cdot\mathcal{W}_1(P,Q). \quad \square\end{aligned} \tag{C.8}$$

□

For $\alpha = 0.95$, the bound is $20\cdot\mathcal{W}_1$, so $\mathcal{W}_1 = 0.01$ guarantees $\text{CVaR}_{0.95}$ agreement within 0.2 units. This provides $\alpha$-agnostic tail regularization without explicit CVaR tuning [69].

### Appendix C.4. Comparison with Alternative Divergences

Table C.7: Theoretical Comparison of Distributional Divergences

| **Property** | $\mathcal{W}_1$ | **KL** | **JS** |
|---|---|---|---|
| Symmetric | ✓ | × | ✓ |
| Triangle inequality (metric) | ✓ | × | × |
| Well-defined for disjoint supports | ✓ | × | × |
| Sensitive to geometry of $\mathbb{R}$ | ✓ | × | × |
| Bounded CVaR gap | ✓ | × | × |
| Parametric sample complexity (1D) | $O(n^{-1/2})$ | Unbounded | Unbounded |

## Appendix D. Sensitivity Analysis

We conduct a sensitivity analysis on five key hyperparameters: the accuracy–profit balance weight $\eta$, sliding window size $T^{\text{in}}$, guidance schedule $\alpha$, learning rate, and PVPNet hidden dimension $d_h$. Table D.8 reports the results.

Table D.8: Hyperparameter Sensitivity Analysis

| **Parameter** | **Setting** | **AR (%)** | **SR** | **MDD (%)** |
|---|---|---|---|---|
| $\eta$ (accuracy–profit balance) | 0.1 (profit-heavy) | 23.5 | 0.81 | 29.1 |
| | 0.3 | 23.1 | 0.80 | 27.8 |
| | 0.5 (balanced) | 22.8 | 0.79 | 27.0 |
| | 0.7 | 21.2 | 0.75 | 26.4 |
| | 0.9 (accuracy-heavy) | 18.6 | 0.66 | 25.1 |
| Sliding window size $T^{\text{in}}$ (days) | 50 | 20.1 | 0.72 | 29.5 |
| | 100 (default) | 22.8 | 0.79 | 27.0 |
| | 150 | 21.9 | 0.77 | 27.8 |
| | 200 | 20.5 | 0.73 | 29.2 |
| $\alpha$ guidance schedule | Fixed 0.3 (explore-heavy) | 19.8 | 0.70 | 28.5 |
| | Fixed 0.7 (exploit-heavy) | 21.5 | 0.76 | 27.3 |
| | Adaptive (default) | 22.8 | 0.79 | 27.0 |
| Learning rate | $10^{-6}$ | 20.9 | 0.74 | 28.1 |
| | $10^{-5}$ (default) | 22.8 | 0.79 | 27.0 |
| | $10^{-4}$ | 21.3 | 0.75 | 28.8 |
| Hidden dimension $d_h$ | 8 | 21.5 | 0.76 | 27.5 |
| | 16 (default) | 22.8 | 0.79 | 27.0 |
| | 32 | 22.1 | 0.78 | 27.2 |

Several observations emerge. **Accuracy–profit balance $\eta$.** Performance is robust across $\eta \in [0.1, 0.7]$, with Sharpe ratios ranging from 0.75 to 0.81. The profit-heavy setting ($\eta = 0.1$) achieves the highest return (23.5%) but also higher drawdown, while the accuracy-heavy setting ($\eta = 0.9$) sacrifices return without commensurate risk reduction, confirming that both objectives contribute positively. **Sliding window size.** The default $T^{\text{in}} = 100$ days performs best; shorter windows (50 days) yield noisier estimates, while longer windows (200 days) reduce adaptability to regime shifts. **Adaptive $\alpha$ guidance.** The adaptive schedule outperforms both fixed exploration-heavy and exploitation-heavy alternatives, validating the design of the buffer-driven annealing mechanism. **Hidden dimension $d_h$.** Increasing $d_h$ from 16 to 32 yields negligible improvement, confirming that 16 dimensions are sufficient for this constrained grammar-based search setting, as the effective branching factor is limited to approximately $|\mathcal{R}| \approx 30$ production rules.

## Appendix E. Additional Discovered Expressions

*Volatility-Adjusted Momentum.*

$$e = \frac{\text{mom}(x_{\text{close}}, 10)}{\text{vol}(x_{\text{close}}, 20)} \times \text{ite}\big(\text{ma}(x_{\text{close}}, 5) > \text{ma}(x_{\text{close}}, 20),\ 1,\ 0\big). \quad \text{(E.1)}$$

*Interpretation:* Momentum normalized by volatility, filtered by trend confirmation (golden cross).

*Mean Reversion with Volume.*

$$e = \big(x_{\text{close}} - \text{ma}(x_{\text{close}}, 20)\big) \times \frac{x_{\text{volume}}}{\text{ma}(x_{\text{volume}}, 20)}. \quad \text{(E.2)}$$

*Interpretation:* Price deviation scaled by relative volume (higher conviction on high volume).